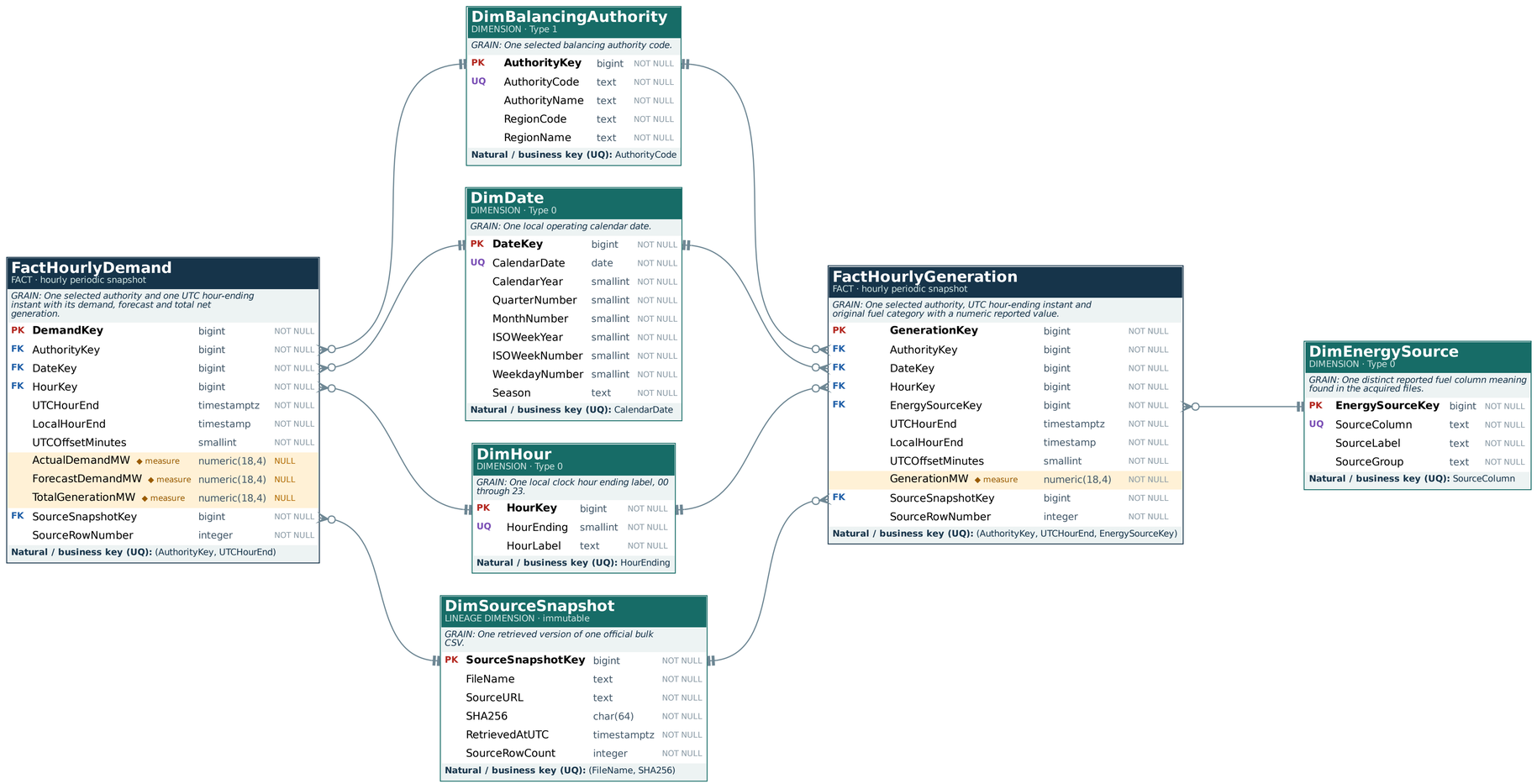
PowerPulse — Physical Dimensional Model

Conformed fact constellation · 5 dimensions + 2 hourly periodic-snapshot facts · PostgreSQL target · DL3004 Phase 1 · FAST NUCES Lahore



How to read **PK** surrogate primary key (warehouse-generated) **FK** foreign key to a dimension PK **UQ** unique natural/business key **◆ measure** (amber rows) fact measure **Crow's foot** = zero-or-many fact rows; double bar = exactly one dimension member (every fact FK is mandatory)

Safe comparison: FactHourlyDemand is unique on (authority, UTC hour); FactHourlyGeneration adds energy source. Aggregate generation to authority + UTC hour *before* joining to demand, so one demand value is never repeated per fuel. Date, Hour, BalancingAuthority and SourceSnapshot are conformed across both facts.